

Study C906289002

Mock Phase 2 Oncology Trial

Protocol Version 1.0 · Synthetic document for demonstration only.

1. Background

Study C906289002 is a Mock Phase 2 oncology trial evaluating investigational agent MX-209 in adults with relapsed or refractory solid tumors who have exhausted standard-of-care therapy. This protocol is provided for demonstration purposes only and does not describe a real clinical investigation; all subjects, sites, sponsors, and outcomes referenced herein are fictional.

Despite advances in targeted therapy and immune checkpoint inhibition over the past decade, a meaningful fraction of patients with advanced solid tumors develop resistance to available treatments and have limited subsequent options. Median overall survival following progression on second-line therapy remains under twelve months across most histologies, and tolerability of cytotoxic salvage regimens is often poor in heavily pretreated populations.

MX-209 is an orally bioavailable small-molecule inhibitor of a tumor-associated kinase implicated in proliferation, DNA damage repair, and acquired resistance to standard cytotoxic therapy. The compound demonstrated single-agent activity across a panel of patient-derived xenograft models spanning non-small cell lung, colorectal, ovarian, and triple-negative breast histologies, with tumor growth inhibition observed at exposures achievable in human dosing.

The first-in-human dose-escalation study (Study C906289001) enrolled 42 subjects across six dose levels. The maximum tolerated dose was not formally reached; the recommended Phase 2 dose (RP2D) of 200 mg once daily was selected on the basis of pharmacokinetic exposure, target engagement biomarkers, and the absence of dose-limiting toxicities at that level. Common treatment-emergent adverse events in the Phase 1 study included fatigue, nausea, transaminase elevation, and neutropenia.

The present study is designed to estimate single-agent efficacy at the RP2D and to further characterize the safety profile in an enriched population. Findings will inform the design of subsequent randomized studies and combination strategies.

2. Objectives

The primary objective of Study C906289002 is to estimate the objective response rate (ORR) of MX-209 monotherapy, defined as the proportion of subjects achieving a confirmed complete or partial response per RECIST v1.1 as assessed by the investigator. The ORR will be summarized with a two-sided 95% confidence interval.

Secondary efficacy objectives include estimation of progression-free survival (PFS), overall survival (OS), duration of response (DoR), and disease control rate (DCR) defined as the proportion of subjects achieving complete response, partial response, or stable disease lasting at least 16 weeks. All efficacy endpoints will be analyzed in the Full Analysis Set.

Secondary safety objectives include characterization of the type, incidence, severity, timing, seriousness, causality, and resolution of treatment-emergent adverse events, with particular attention to adverse events of special interest (AESIs) as defined in Section 6. Dose modifications, treatment interruptions, and discontinuations due to toxicity will be tabulated.

Exploratory objectives include characterization of the pharmacokinetic profile at the RP2D under steady-state conditions, assessment of candidate predictive biomarkers including baseline circulating tumor DNA mutational status and on-treatment dynamics, and exploration of resistance mechanisms through paired tumor biopsies collected at baseline and at the time of progression where clinically feasible.

3. Study Design

This is a Phase 2, open-label, single-arm, multicenter study evaluating MX-209 monotherapy in adults with relapsed or refractory advanced solid tumors. Approximately 80 subjects will be enrolled across 12 investigative sites in North America and Europe. The study consists of three periods: a screening period of up to 28 days, an open-label treatment period of repeating 28-day cycles, and a follow-up period extending to 30 days after the last dose for safety and quarterly thereafter for survival.

Subjects will receive MX-209 at the RP2D of 200 mg orally once daily on a continuous schedule. Treatment will continue until radiographic progression, unacceptable toxicity, withdrawal of consent, investigator decision, or completion of two years of therapy, whichever occurs first. Subjects who derive clinical benefit at the two-year mark may continue treatment with sponsor agreement.

Disease assessments by computed tomography or magnetic resonance imaging will be performed at screening and every eight weeks (every two cycles) for the first twelve months, and every twelve weeks thereafter, until documented progression. Responses must be confirmed by a repeat assessment at least four weeks after the initial observation.

Safety assessments including vital signs, ECOG performance status, physical examination, hematology, serum chemistry, coagulation, and 12-lead ECG will be performed at the start of each cycle and as clinically indicated. Concomitant medications will be recorded at every visit. Quality of life will be assessed at screening, every other cycle, and at end of treatment using the EORTC QLQ-C30 instrument.

Translational research samples — peripheral blood for circulating tumor DNA and pharmacokinetics, and optional tumor biopsies — will be collected per the schedule in Appendix C. Pharmacokinetic sampling will be performed on Cycle 1 Day 1, Cycle 1 Day 15, and Cycle 2 Day 1.

4. Eligibility

To be eligible, subjects must be at least 18 years of age at the time of consent, have a histologically or cytologically confirmed advanced solid tumor of any histology, have radiographically measurable disease per RECIST v1.1, and have documented progression on or intolerance to at least one prior line of systemic therapy for advanced or metastatic disease.

Subjects must have an ECOG performance status of 0 or 1, a life expectancy of at least 12 weeks in the investigator's judgment, and provide written informed consent in accordance with local regulatory and institutional review board requirements. Women of childbearing potential must have a negative serum pregnancy test within seven days of the first dose and agree to use highly effective contraception for the duration of treatment and for 90 days following the last dose.

Adequate organ function must be documented within 14 days prior to first dose: absolute neutrophil count at least $1.5 \times 10^9/L$, platelet count at least $100 \times 10^9/L$, hemoglobin at least 9.0 g/dL without transfusion in the prior two weeks, total bilirubin at most 1.5 x upper limit of normal (3 x ULN in subjects with documented Gilbert's syndrome), AST and ALT at most 2.5 x ULN (5 x ULN in subjects with documented hepatic involvement of malignancy), and creatinine clearance at least 50 mL/min by Cockcroft-Gault estimation.

Key exclusion criteria include known active central nervous system metastases (treated and stable CNS disease without ongoing corticosteroid requirement is permitted following at least 28 days of clinical stability), known leptomeningeal involvement, uncontrolled intercurrent illness including but not limited to symptomatic congestive heart failure, unstable angina, myocardial infarction within six months, clinically significant arrhythmia, or active uncontrolled infection requiring systemic therapy.

Additional exclusions include receipt of any investigational agent within 28 days or five half-lives (whichever is shorter) prior to the first dose, major surgery within 28 days, palliative radiotherapy to a non-target lesion within 14 days, prior allogeneic stem cell or solid organ transplant, history of another malignancy within three years (with the exception of adequately treated non-melanoma skin cancer or carcinoma in situ), pregnancy or breastfeeding, and any condition that in the investigator's judgment would interfere with study participation or compromise the interpretation of study data.

5. Dosing

MX-209 will be supplied as 50 mg and 100 mg film-coated tablets. The recommended Phase 2 dose is 200 mg administered orally once daily on a continuous schedule. Doses should be taken at approximately the same time each day, with or without food, and swallowed whole with water. Tablets must not be crushed, chewed, or dissolved.

A missed dose may be taken within six hours of the scheduled administration time; if more than six hours have elapsed, the dose should be skipped and the next dose taken at the scheduled time the following day. Subjects must not take a double dose to compensate for a missed dose. Vomiting within 30 minutes of administration may be followed by a single re-dose; vomiting beyond that window should not be re-dosed.

Two dose reductions are permitted in response to treatment-emergent toxicity, to 150 mg and then to 100 mg, in accordance with the dose modification schema in Appendix B. Subjects requiring a third dose reduction will be discontinued from study treatment. Dose re-escalation is not permitted once a reduction has occurred.

Dose interruptions for non-hematologic toxicity should generally not exceed 21 consecutive days; interruptions of longer duration will result in treatment discontinuation unless explicit sponsor concurrence is documented. Hematologic toxicity may permit interruptions of up to 28 days at investigator discretion.

Drug accountability records must be maintained at each site. Subjects are instructed to return all unused tablets and empty containers at each on-site visit. Compliance will be assessed by tablet count and will be documented in the source record. Subjects with compliance below 75% over two consecutive cycles will be reviewed by the medical monitor.

Concomitant strong inhibitors and inducers of CYP3A4 are prohibited from screening through 30 days after the last dose. Moderate inhibitors and inducers are permitted only when clinically necessary and require dose monitoring per Appendix D.

6. Adverse Event Definitions

An adverse event (AE) is any untoward medical occurrence in a subject administered the investigational product, whether or not considered related to the product. This definition includes new signs, symptoms, or diseases, as well as worsening of pre-existing conditions. AEs will be graded according to NCI CTCAE v5.0 and coded using the latest version of MedDRA available at the time of database lock.

A serious adverse event (SAE) is any AE that meets one or more of the following criteria: results in death; is life threatening (i.e., places the subject at immediate risk of death from the event as it occurred); requires inpatient hospitalization or prolongation of existing hospitalization; results in persistent or significant disability or incapacity; is a congenital anomaly or birth defect in the offspring of a study subject; or is an important medical event that may jeopardize the subject or may require medical or surgical intervention to prevent one of the above outcomes, based on medical judgment.

Causality assessment will be performed by the investigator for every AE using a five-category scale: not related, unlikely related, possibly related, probably related, and definitely related. The Sponsor's medical monitor will perform an independent causality assessment for all SAEs. In the event of discordance, both assessments will be retained in the safety database and the more conservative assessment will be used for expedited regulatory reporting.

Severity grading per CTCAE v5.0: Grade 1 (mild; asymptomatic or mild symptoms, clinical or diagnostic observations only, intervention not indicated), Grade 2 (moderate; minimal, local, or non-invasive intervention indicated), Grade 3 (severe or medically significant but not immediately life threatening; hospitalization or prolongation indicated), Grade 4 (life threatening; urgent intervention indicated), and Grade 5 (death related to AE).

Adverse events of special interest (AESIs) for this protocol, requiring expedited internal review even when not meeting SAE criteria, include: febrile neutropenia of any grade; Grade 3 or higher hepatic enzyme elevation (AST, ALT, total bilirubin) including any case meeting Hy's law criteria; any infection of Grade 3 or higher severity; Grade 3 or higher cardiac arrhythmia; QTc prolongation to greater than 500 ms or by more than 60 ms from baseline; interstitial lung disease or pneumonitis of any grade; and any Grade 4 or 5 hematologic toxicity.

Pregnancies occurring in a study subject or in the partner of a male subject during treatment or within 90 days after the last dose must be reported to the Sponsor within 24 hours of investigator awareness. Pregnancy outcomes will be followed regardless of whether the pregnancy was discovered during or after treatment.

7. Safety Monitoring

All subjects will be monitored continuously for adverse events from the time of first dose through 30 days after the last dose of study drug. Investigators will assess and grade events at each scheduled visit and as clinically indicated, and will document onset, severity, seriousness, causality, action taken with study drug, concomitant interventions, and outcome.

Serious adverse events must be reported to the Sponsor's pharmacovigilance unit within 24 hours of investigator awareness, regardless of causality assessment or relationship to study drug. SAE reporting is performed on the designated SAE form and submitted via the validated electronic reporting system; paper submission is permitted only in the event of system unavailability and must be followed by electronic entry within 48 hours.

The Sponsor will forward qualifying expedited SAEs to applicable regulatory authorities and to all participating investigators in accordance with ICH E2A and local expedited reporting requirements. The Sponsor will also distribute periodic Development Safety Update Reports (DSURs) to investigators and regulatory authorities on the schedule defined in the safety management plan.

An independent Data Monitoring Committee (DMC) consisting of three external members — two clinical experts in medical oncology and one biostatistician — will review aggregate safety data at pre-specified intervals (after the first 20 subjects have completed two cycles, after the first 40 subjects have completed four cycles, and quarterly thereafter). The DMC may recommend protocol modifications, dose adjustments, enrollment pause, or study suspension based on emerging safety signals.

Pre-specified stopping rules include: more than 25% of enrolled subjects experiencing a Grade 4 or 5 treatment-related adverse event; any single treatment-related Grade 5 event meeting Hy's law criteria; or two or more treatment-related Grade 5 events of any cause. Triggering any stopping rule will result in immediate enrollment pause pending DMC review.

Each site is responsible for local reporting to its Institutional Review Board or Ethics Committee in accordance with local requirements. Sites must also ensure that all study personnel involved in safety assessment have completed current Good Clinical Practice training and protocol-specific training prior to enrolling subjects.

8. Statistical Analysis

The primary analysis population for efficacy is the Full Analysis Set (FAS), defined as all enrolled subjects who receive at least one dose of MX-209. The Safety Analysis Set is identical to the FAS for this protocol. The Per-Protocol Set, defined as FAS subjects without major protocol deviations, will be used for sensitivity analyses of the primary endpoint.

The planned sample size of approximately 80 subjects provides adequate precision for estimating the primary endpoint. Assuming a true ORR of 25%, a sample size of 80 yields a two-sided 95% Clopper-Pearson confidence interval of approximately 16% to 36%, which is considered acceptable for inferring clinical benefit relative to a null rate of 10% based on historical performance of available salvage therapies.

Objective response rate will be summarized with a two-sided 95% Clopper-Pearson exact confidence interval. Time-to-event endpoints (PFS, OS, DoR) will be summarized using Kaplan-Meier methodology, with median estimates and corresponding two-sided 95% confidence intervals computed using the Brookmeyer-Crowley method. Disease control rate will be summarized with a two-sided 95% Clopper-Pearson confidence interval. No formal hypothesis testing is planned in this single-arm study.

Adverse events will be summarized by MedDRA system organ class and preferred term, by maximum CTCAE grade per subject, and by relationship to study drug. Treatment-emergent AEs are defined as any AE with onset on or after the first dose of MX-209 through 30 days after the last dose. Laboratory abnormalities will be summarized by maximum post-baseline grade and by shift tables relative to baseline grade. Concomitant medications will be coded using the WHO Drug Dictionary.

Missing data for the primary endpoint will be handled by defining non-evaluable subjects as non-responders in the primary analysis. Sensitivity analyses will exclude non-evaluable subjects from the denominator. For time-to-event endpoints, subjects without an event at the analysis cutoff will be censored at the date of last adequate disease assessment.

An interim analysis is planned after the first 40 enrolled subjects have completed at least four cycles of treatment, for the purpose of supporting regulatory interactions and informing potential expansion cohorts. The interim analysis is descriptive only and will not include formal stopping criteria for efficacy or futility; the study will continue to its planned total enrollment regardless of interim results.